

Multimedia Computing

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Analog Video

- ▶ Represents video as a continuous signal. The information about color and brightness is encoded in varying electrical voltages.
- ▶ The video signal is typically transmitted via **composite video, component video, or S-video cables**.
- ▶ Examples include older formats like VHS, Betacam, and analog television broadcasts (NTSC, PAL, SECAM).

Analog Video

- ▶ Transmits video over short distances without compression, which can result in degradation of quality due to noise, interference, and signal attenuation.
- ▶ Requires physical media like videotapes for storage. Over time, the quality of the video degrades due to wear and aging of the media.

Types of Video signals

- ▶ **Component video**
- ▶ Higher-end video systems make use of three separate video signals for the red, green, and blue image planes. Each color channel is sent as a separate video signal.
 - a) Most computer systems use component video, with separate signals for R,G and B signals.
 - b) For any color separation scheme, Component Video gives the best color reproduction since there is no “crosstalk” between the three channels
 - c) This is not the case for S-Video or composite video. Component video, however, require more bandwidth and good synchronization of the three components.

Composite Video

- ▶ **Composite video:** color(“chrominance”) and intensity(“luminance”) signals are mixed into a single carrier wave.
 - a) Chrominance is a composition of two color components (I and Q or U and V).
 - b) In NTSC TV. I and Q are combined into a chroma signal and a color subcarrier is then employed to put the chroma signal at the high-frequency end of the signal shared with the luminance signal.
 - c) The chrominance and luminance components can be separated at the receiver end and then the two color components can be further recovered.
 - d) When connecting to TVs or VCR, composite video uses only one wire and video color signals are mixed, not sent separately. The audio and sync signals are addition to this one signal.

S-Video - Signals

- ▶ S-Video: as a compromise (separate video or super video, e.g. in S-VHS) uses **two wires one for luminance and another for a composite chrominance signal**.
- ▶ As a result there is **less crosstalk between the color information** and the crucial gray-scale information.
- ▶ The reason for placing luminance into its own part of the signal is that black and white information is most crucial for visual perception.

-In fact, humans are able to differentiate spatial resolution in grayscale images with a much higher acuity than for the color part of color images.

- As a result, we can send less accurate color information than must be sent for intensity information. -we can only see fairly large blobs of color, so it makes sense to send less color details.

Analog Video

- ▶ Analog video represented as a continuous (time-varying) signal. Up until last decade, most TV programs were sent and received as an analog signal. Once the electrical signal is received, we may assume that brightness is at least a monotonic function of voltage, if not necessarily linear, because of gamma correction

Analog Video

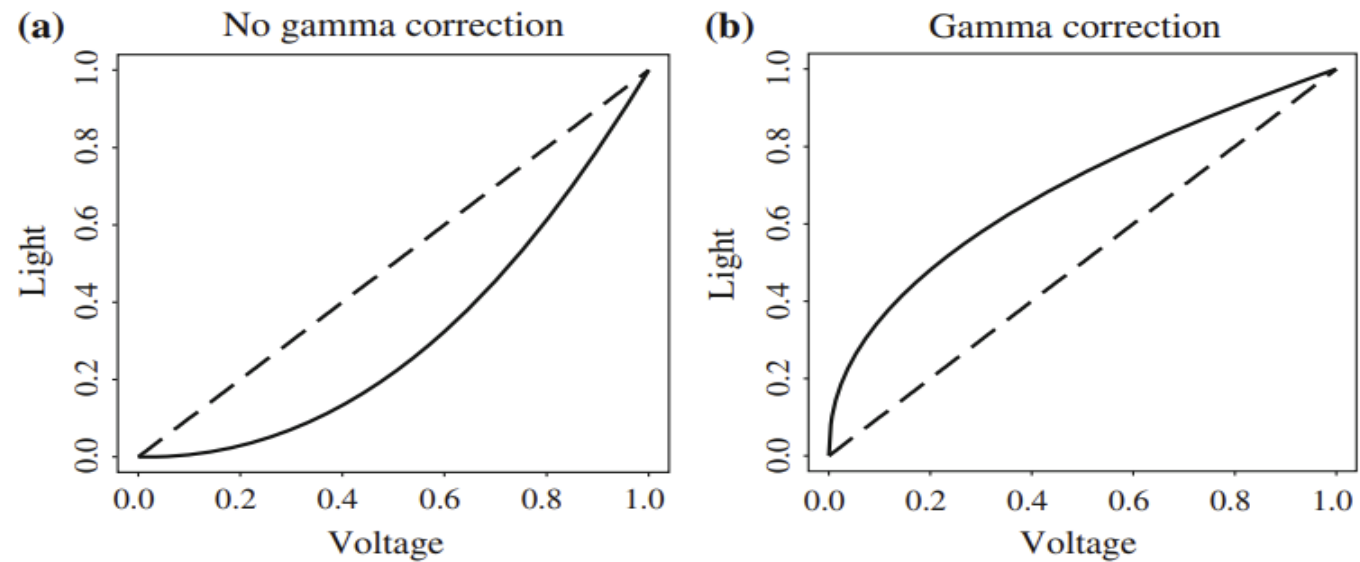


Fig. 4.6 **a** Effect of putative standard CRT (mimicked by an actual modern display) on light emitted from screen (voltage is normalized to range 0..1). **b** Gamma correction of signal

Analog Video

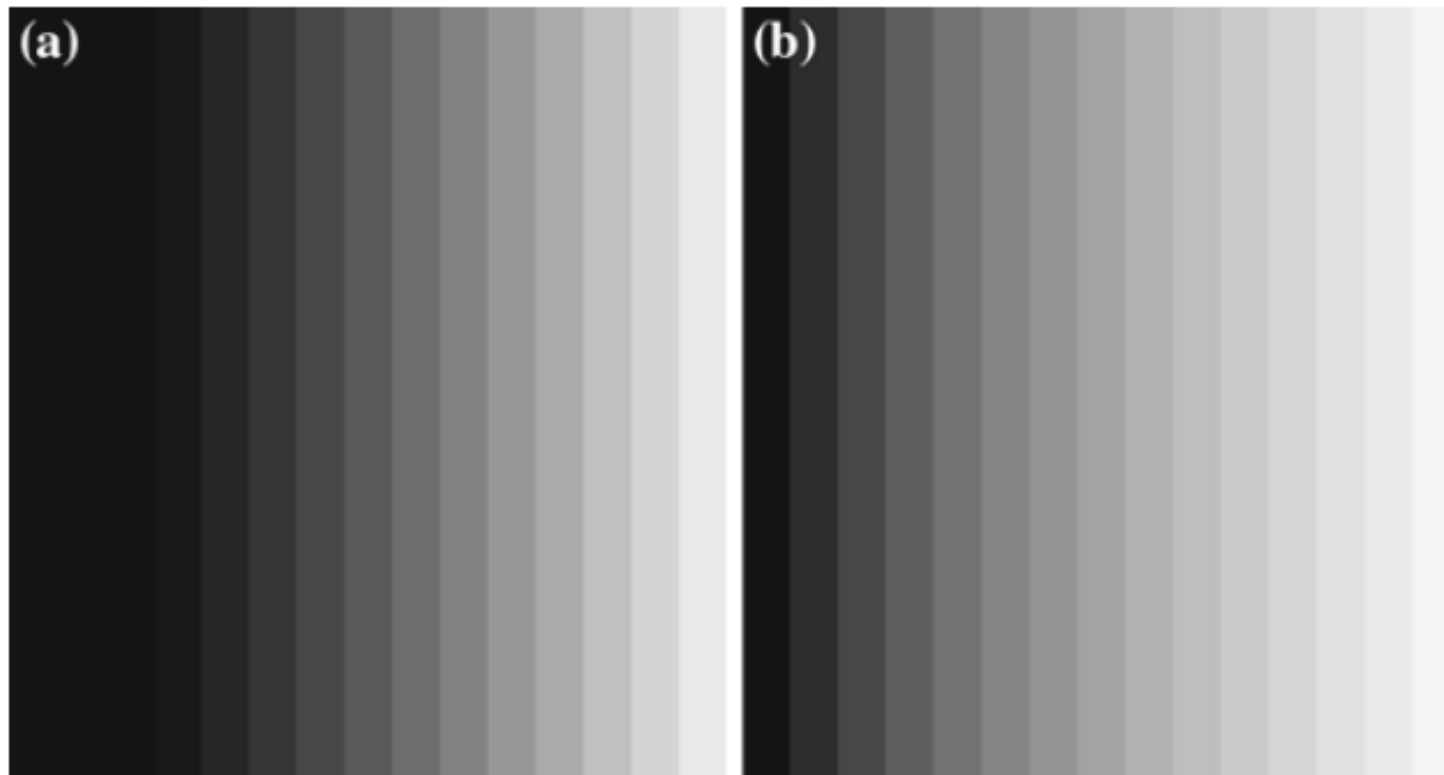


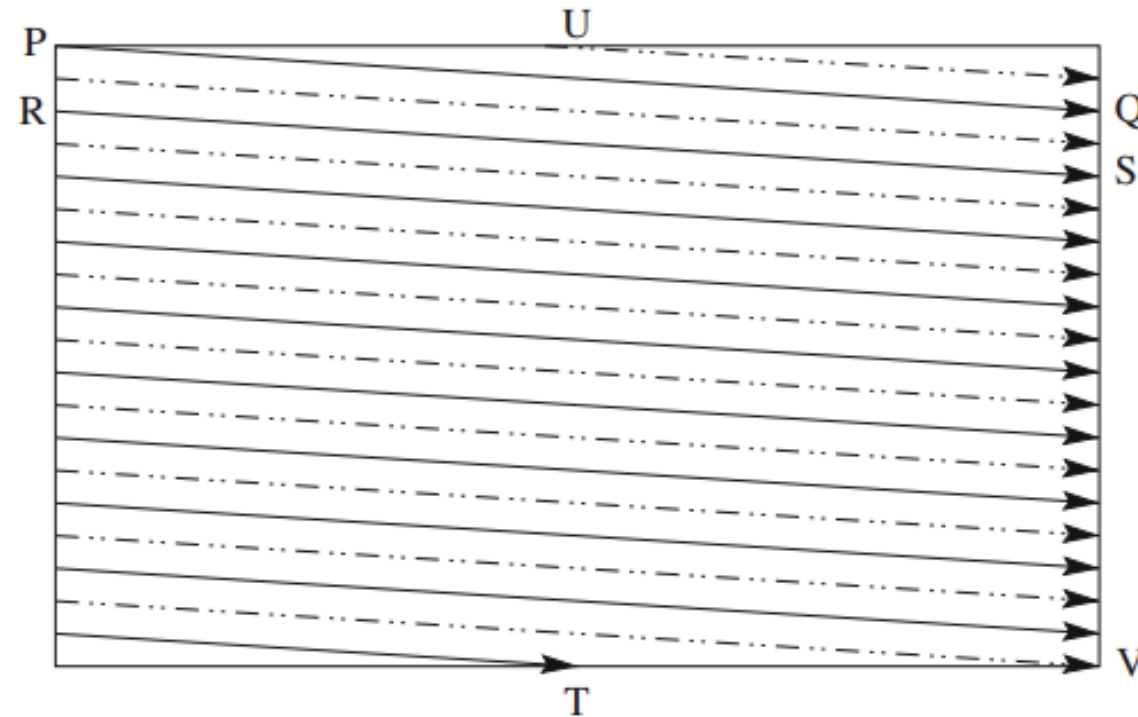
Fig. 4.7 **a** Display of ramp from 0 to 255, with no gamma correction. **b** Image with gamma correction applied

Analog Video

- ▶ An analog signal $f(t)$ samples a time-varying image. So-called progressive scanning traces through a complete picture (a frame) row-wise for each time interval. A high-resolution computer monitor typically uses a time interval of $1/72$ s.
- ▶ In TV and in some monitors and multimedia standards, another system, interlaced scanning, is used. Here, the odd-numbered lines are traced first, then the even-numbered lines. This results in “odd” and “even” fields—two fields make up one frame.

Analog Video

Fig. 5.1 Interlaced raster scan



- ▶ Figure 5.1 shows the **scheme used**. First the **solid (odd) lines** are traced **P to Q**, then R to S etc. ending at T; then the even field starts at U and ends at V.
- ▶ The jump from Q to R etc. in Figure 5.1 is called the **horizontal retrace**, during which the electronic beam in the CRT is blanked. The jump from T to U or **V to P** is called **vertical retrace**.

Analog Video

- ▶ Because of interlacing, the odd and even lines are displaced in time from each other- generally not noticeable except when very fast action is taking place on screen, when blurring may occur.
- ▶ For example, in the video in fig.5.2 the moving helicopter is blurred more than is the still background.

Analog Video

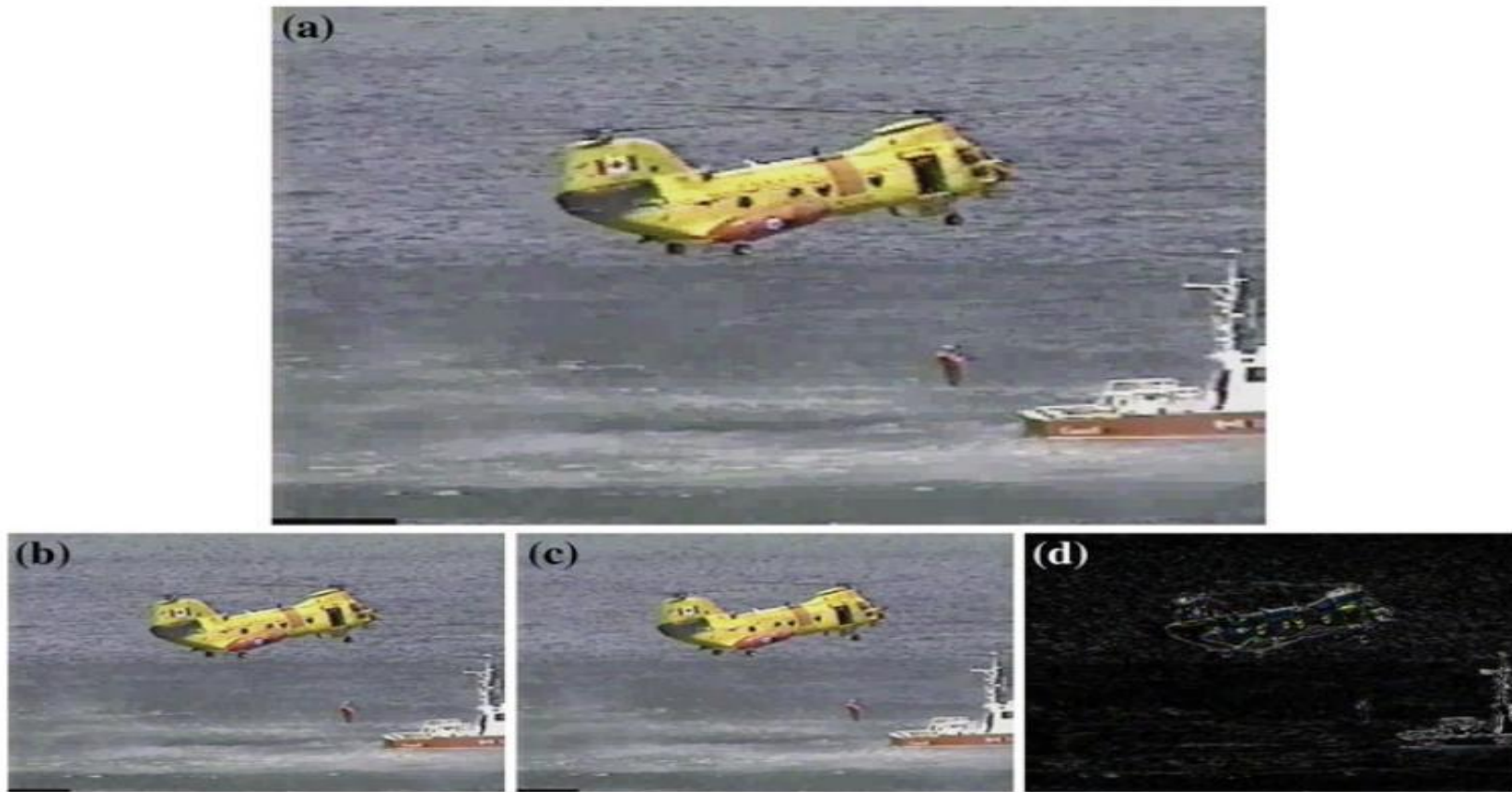


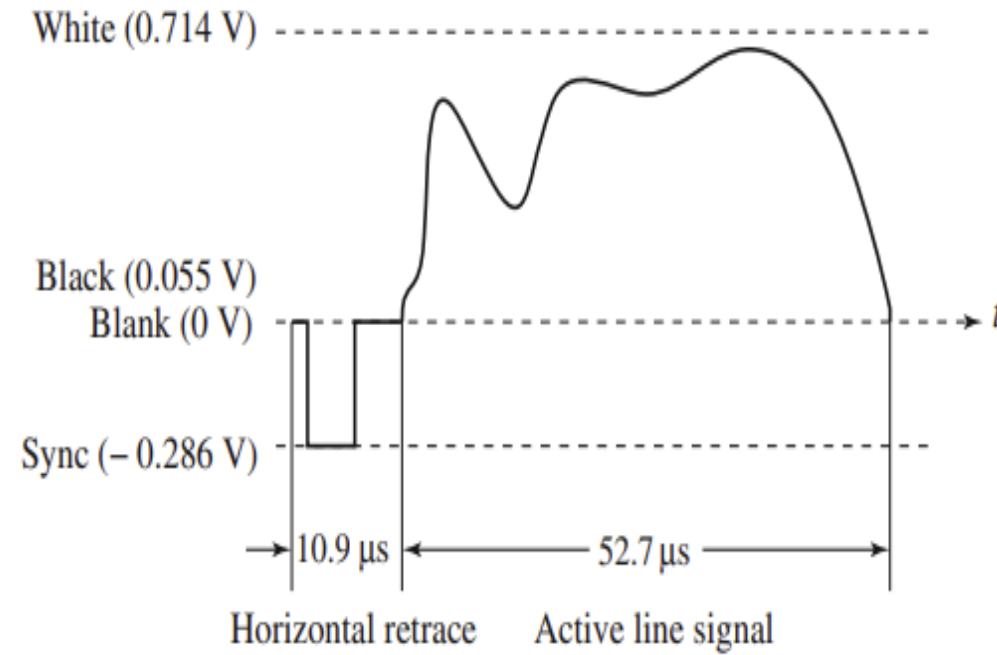
Fig. 5.2 Interlaced scan produces two fields for each frame: **a** the video frame; **b** Field 1; **c** Field 2; **d** difference of fields

Analog Video

- Since it is sometimes necessary to change the frame rate, resize, or even produce stills from an interlaced source video, various schemes are used to “de-interlace” it.
 - a) The simplest de-interlacing method consists of discarding one field and duplicating the scan lines of the other field. The information in one field is lost completely using this simple technique.
 - b) Other more complicated methods that retain information from both fields are also possible.
- Analog video use a small voltage offset from zero to indicate “black”, and another value such as zero to indicate the start of a line. For example, we could use a “blacker-than-black” zero signal to indicate the beginning of a line.

NTSC Video

Fig. 5.3 Electronic signal for one NTSC scan line



NTSC Video

NTSC (National Television System Committee) TV standard is mostly used in North America and Japan. It uses the familiar 4:3 **aspect ratio** (i.e., the ratio of picture width to its height) and uses 525 scan lines per frame at 30 frames per second (fps).

- a) NTSC follows the interlaced scanning system, and each frame is divided into two fields, with 262.5 lines/field.
- b) Vertical retrace takes place during 20 lines reserved for control information at the beginning of each field.
- c) Similarly, almost 1/6 of the raster at the left side is blanked for horizontal retrace and sync. The non-blanking pixels are called *active pixels*.
- d) It is known that pixels often fall in-between the scan lines. Therefore, even with non-interlaced scan, NTSC TV is only capable of showing about 70% of the active lines. With interlaced scan, this could be as low as 50%.

PAL video

- **PAL (Phase Alternating Line)** is a TV standard widely used in Western Europe, China, India, and many other parts of the world.
- PAL uses 625 scan lines per frame, at 25 frames/second, with a 4:3 aspect ratio and interlaced fields.
 - (a) PAL uses the YUV color model. It uses an 8 MHz channel and allocates a bandwidth of 5.5 MHz to Y, and 1.8 MHz each to U and V. The color subcarrier frequency is $f_{sc} \approx 4.43$ MHz.
 - (b) In order to improve picture quality, chroma signals have alternate signs (e.g., +U and -U) in successive scan lines, hence the name “Phase Alternating Line”.
 - (c) This facilitates the use of a (line rate) comb filter at the receiver — the signals in consecutive lines are averaged so as to cancel the chroma signals (that always carry opposite signs) for separating Y and C and obtaining high quality Y signals.

SECAM video

- ▶ SECAM stands for system Electronique couleur Avec Memoire , the third major broadcast TV standard.
- ▶ SECAM also uses 625 lines per frame, at 25 frames per second, with a 4:3 aspect ratio and interlaced fields.
- ▶ SECAM and PAL are very similar. They differ slightly in their color subcarriers at 4.25MHz and 4.41 MHz respectively.
- ▶ They are sent in alternate lines i.e only one of the U or V signals will be sent on each scan line.

Comparison of Analog Broadcast TV systems

TV System	Frame Rate (fps)	# of Scan Lines	Total Channel Width (MHz)	Bandwidth Allocation (MHz)		
				Y	I or U	Q or V
NTSC	29.97	525	6.0	4.2	1.6	0.6
PAL	25	625	8.0	5.5	1.8	1.8
SECAM	25	625	8.0	6.0	2.0	2.0

Digital video

- The advantages of digital representation for video are many. For example:
 - (a) Video can be stored on digital devices or in memory, ready to be processed (noise removal, cut and paste, etc.), and integrated to various multimedia applications;
 - (b) Direct access is possible, which makes nonlinear video editing achievable as a simple, rather than a complex, task;
 - (c) Repeated recording does not degrade image quality;
 - (d) Ease of encryption and better tolerance to channel noise.

Chroma Subsampling

- ▶ Since humans see color with much less spatial resolution than they see black and white, it makes sense to “decimate” the chrominance signal.
- ▶ Numbers are given stating how many pixel values, per four original pixels are actually sent:
 - a) The chroma subsampling scheme “4:4:4” indicates that no chroma subsampling is used : each pixels Y, Cb and Cr values are transmitted 4 for each of Y, Cb, Cr.

Chroma Subsampling

- (b) The scheme “4:2:2” indicates horizontal subsampling of the Cb, Cr signals by a factor of 2. That is, of four pixels horizontally labelled as 0 to 3, all four Ys are sent, and every two Cb’s and two Cr’s are sent, as (Cb0, Y0)(Cr0, Y1)(Cb2, Y2)(Cr2, Y3)(Cb4, Y4), and so on (or averaging is used).
- (c) The scheme “4:1:1” subsamples *horizontally* by a factor of 4.
- (d) The scheme “4:2:0” subsamples in both the *horizontal* and *vertical* dimensions by a factor of 2. Theoretically, an average chroma pixel is positioned between the rows and columns as shown Fig.5.6.

Chroma Subsampling



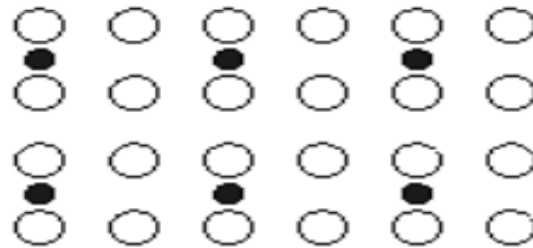
4:4:4



4:2:2



4:1:1



4:2:0

- Pixel with only Y value
- Pixel with only Cr and Cb values
- ⦿ Pixel with Y, Cr, and Cb values

HDTV (High Definition TV)

- The main thrust of **HDTV** (High Definition TV) is not to increase the “definition” in each unit area, but rather to increase the visual field especially in its width.
- For video, MPEG-2 is chosen as the compression standard. For audio, AC-3 is the standard. It supports the so-called 5.1 channel Dolby surround sound, i.e., five surround channels plus a subwoofer channel.
- The salient difference between conventional TV and HDTV:
 - (a) HDTV has a much wider aspect ratio of 16:9 instead of 4:3.
 - (b) HDTV moves toward progressive (non-interlaced) scan. The rationale is that interlacing introduces serrated edges to moving objects and flickers along horizontal edges.

HDTV (High Definition TV)

- It is planned to replace all analog broadcast services with digital TV broadcasting by the year 2009. The services provided will include:
 - **SDTV (Standard Definition TV)** : the current NTSC TV or higher.
 - **EDTV (Enhanced Definition TV)** : 480 active lines or higher, i.e., the third and fourth rows in Table 5.4.
 - **HDTV (High Definition TV)** : 720 active lines or higher.

Video capture

```
import cv2

# Open a connection to the webcam (0 is the default webcam)
cap = cv2.VideoCapture(0)

# Check if the webcam is opened correctly
if not cap.isOpened():
    print("Error: Could not open webcam.")
    exit()

# Capture video frames in a loop
while True:
    # Capture frame-by-frame
    ret, frame = cap.read()
    # If frame is read correctly, ret is True
    if not ret:
        print("Error: Could not read frame.")
        break
```

Video capture

```
# Display the captured frame
```

```
cv2.imshow('Webcam Video', frame)
```

```
# Break the loop if 'q' is pressed
```

```
if cv2.waitKey(1) & 0xFF == ord('q'):
```

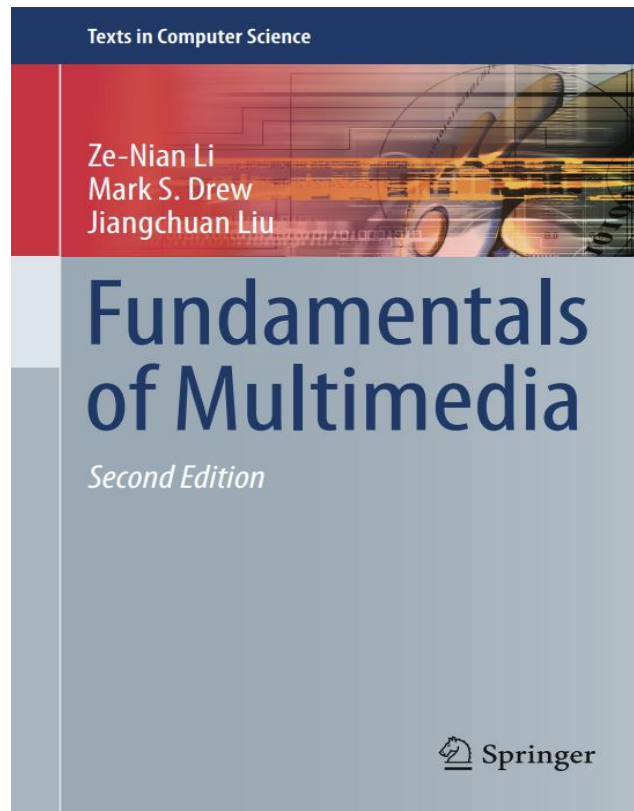
```
    break
```

```
# Release the webcam and close all OpenCV windows
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

Reference



Question



Thank you

